



High Resolution Imaging of the Intervertebral Disc Degeneration using EPIC-microCT

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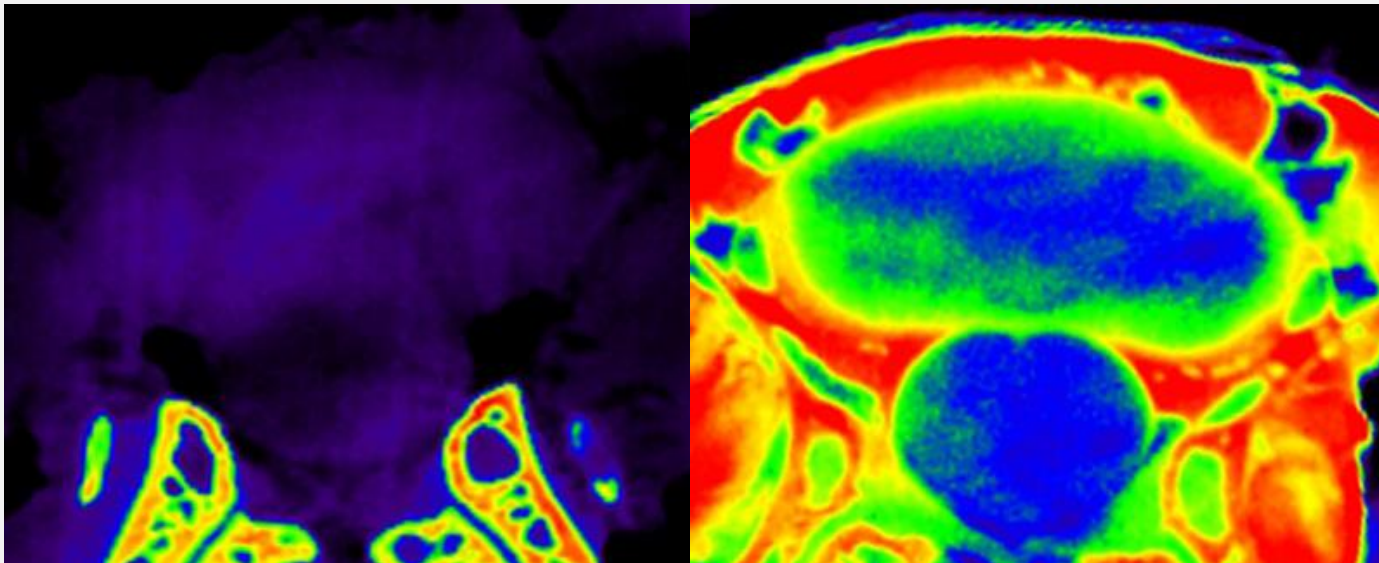
Introduction

- By the age of 50, 80% of the population will show some signs of degenerative disc disease
- Intervertebral disc degeneration, characterized by the loss of glycosaminoglycans (GAGs) from the matrix, causes back pain, instability, and can lead to serious complications
- Lack of high resolution, fast, and cost-effective methods for characterizing disc degradation and regeneration in small animal models for disc disease is a limiting factor in the development of regenerative treatments for degenerative disc disease
- EPIC-microCT is a recently developed CT technique which has demonstrated the ability to provide high resolution characterization of articular cartilage in rats, mice and rabbits
- The goal of our project is to adapt EPIC-microCT to intervertebral disc analysis and assess its ability to effectively characterize disc degradation in a rabbit model

EPIC-microCT technique

EPIC-microCT (Equilibrium Partitioning of an Ionic Contrast agent) is a high resolution imaging technique based on the following principles:

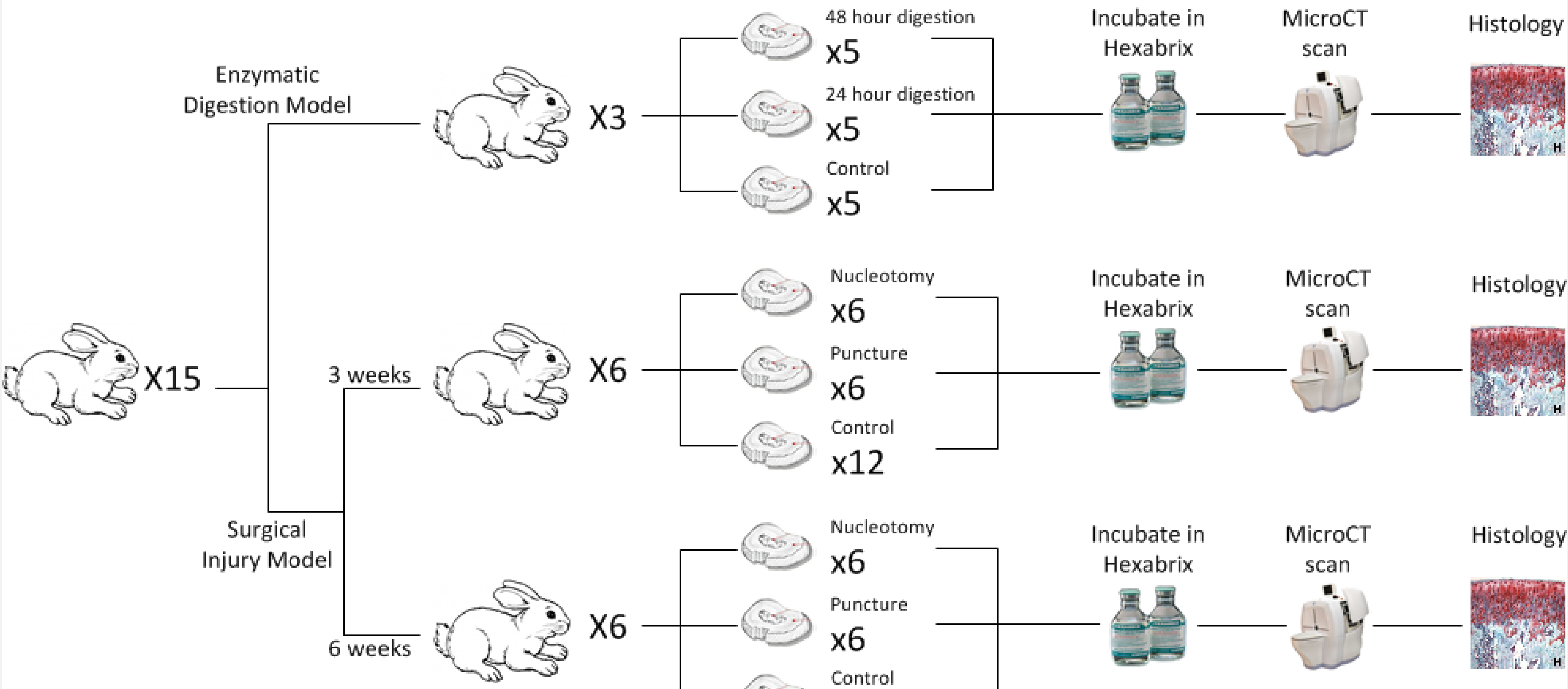
- Contrast agents allows for visualization of soft tissue using x-ray based CT imaging
- The negatively charged contrast agent Hexabrix (Guerbet, Inc.) will be repelled by the negatively charged glycosaminoglycans (GAGs)
- Hexabrix contrast-enhanced tissue will trend towards an increase in CT signal in areas of decreasing GAG content.
- CT signal can then be used as an indirect measurement of GAG content
- As intervertebral disc GAG content is the key marker in disc degradation, EPIC-microCT should be sensitive to degenerative and regenerative changes in the intervertebral disc



False color axial CT slices of intervertebral discs showing contrast enhancement with Hexabrix: non-contrast enhanced (left) and after 24 hour incubation in 50% Hexabrix contrast agent (right).

Modeling Disc Degradation

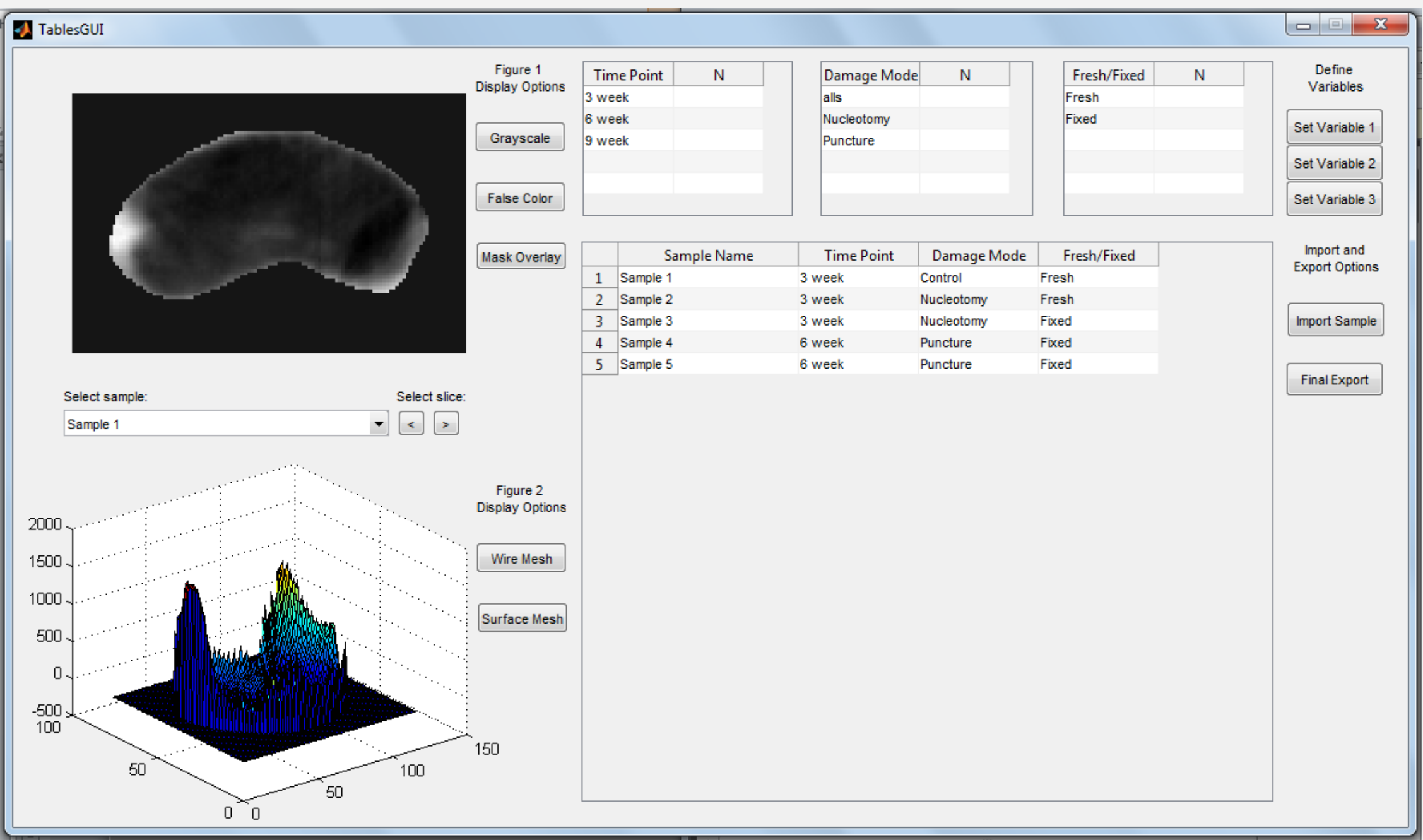
- In vitro GAG digestion was achieved via injection of the enzyme papain into the nucleus pulposus (the gelatinous center of the disc where the majority of the GAG content resides)
- Surgical degradation was induced by annular puncture to model gradual degradation and nucleotomy to model severe and traumatic degradation
- GAG content of each sample was assessed using EPIC-microCT and histology (the current gold standard) for comparison



(Rabbit picture courtesy supercoloring.com, intervertebral disc adapted from cavalierhealth.org, bottles courtesy inmedicine.com, microCT courtesy Gamma Medica, histology courtesy Xie et al [3])

MATLAB Analysis

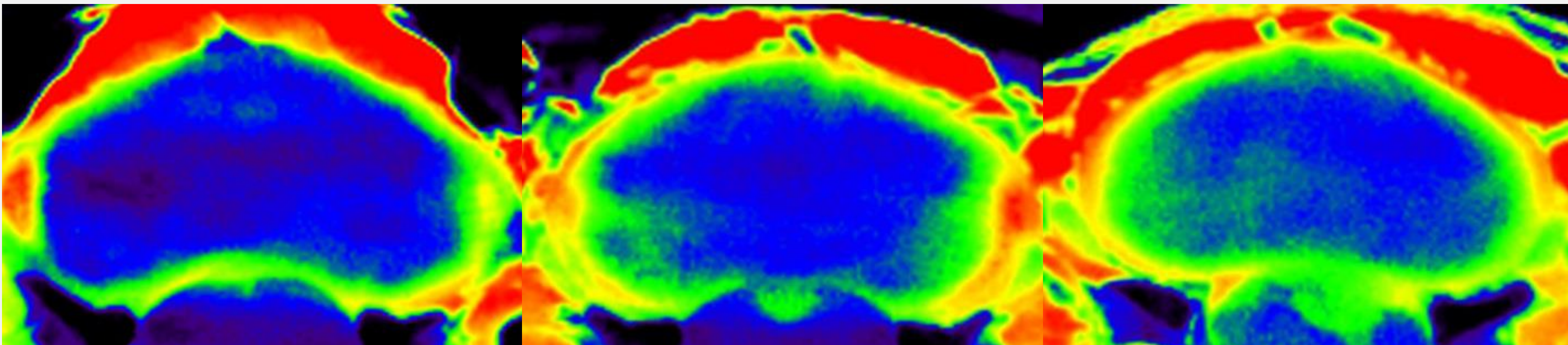
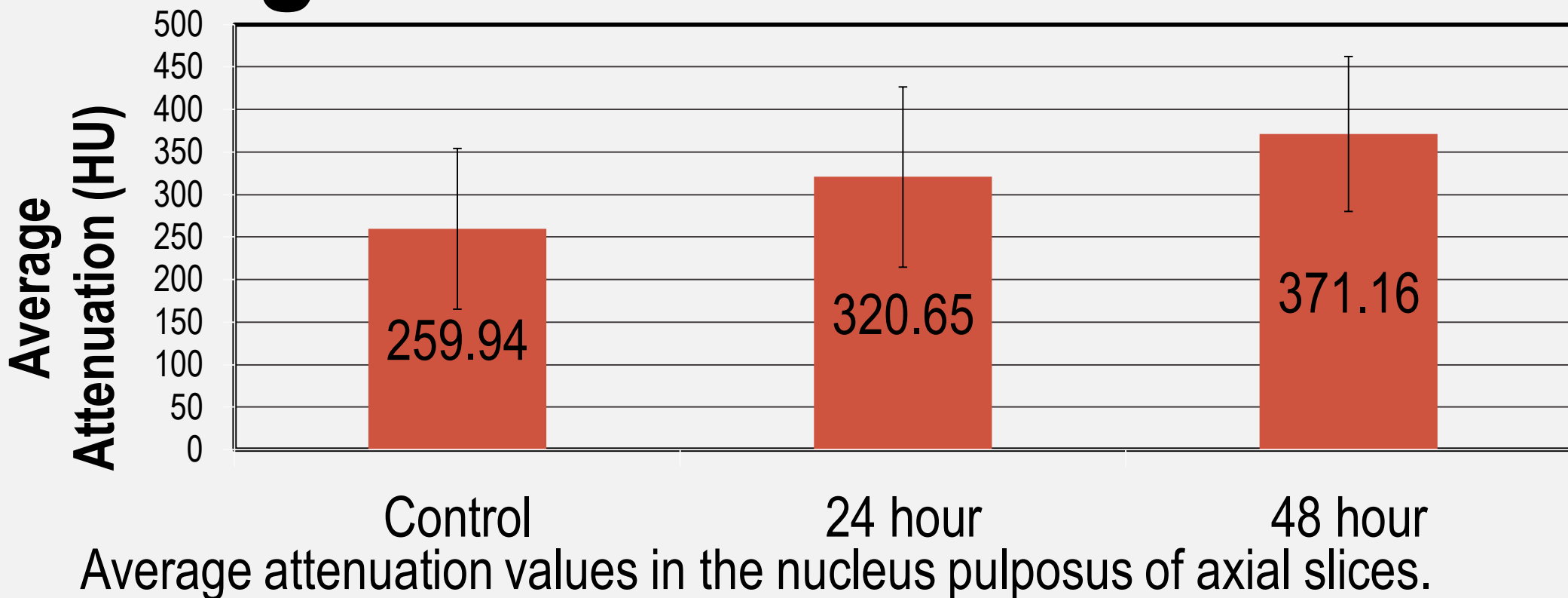
MATLAB analysis of raw CT data was analyzed through a developed and implemented graphical user interface.



Key features:

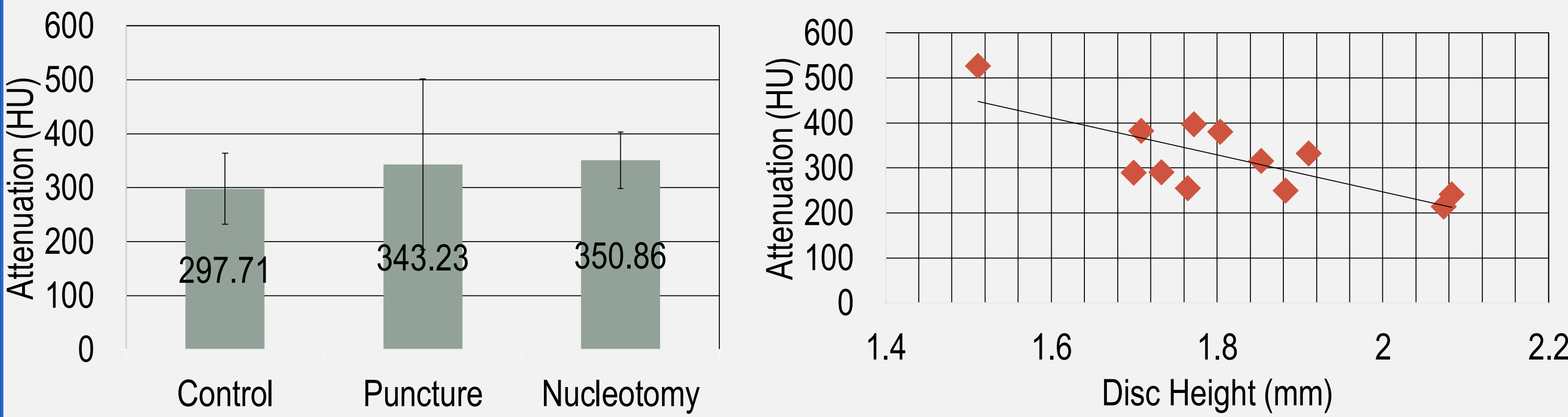
- Manual disc outlining
- Nucleus pulposus Isolation for increased GAG sensitivity
- Organization of scans by variable group, up to 3 variables
- Intuitive generation and manipulation of figures
- Convenient data export for easy analysis

Enzymatic Digestion Model



False color axial CT slices of contrast-enhanced discs showing increasing attenuation with digestion: healthy control disc (left), after 24 hours of papain digestion (center), and after 48 hours of papain digestion (right).

Sugically-Induced Degeneration Model



Average attenuation values of whole disc axial slices.

Correlation between whole disc attenuation and disc height ($R^2 = 0.5676$).

Average attenuation of nucleus pulposus normalized to controls was 1.33 ± 0.41 in nucleotomy and 1.09 ± 0.48 in the puncture model.

Discussion

EPIC-microCT clearly demonstrates the ability to provide contrast enhancement to the intervertebral disc and that the contrast agent is able to diffuse into the center effectively. The enzymatic digestion model of disc degradation demonstrates increased attenuation with increased papain digestion, suggesting an inverse relationship between GAG content and attenuation. The surgically-induced degradation model demonstrates increases in attenuation associated with puncture and nucleotomy at 3 weeks. A correlation between decreasing disc height, a marker of in vivo disc degradation, and increasing attenuation is also observed. Finally, higher attenuation values are observed in the nucleus pulposus as compared to the rest of the disc. The 6-week surgical model and histological analyses of all models are currently underway. The difference between healthy and damaged discs is expected to be more pronounced in the longer 6-week time point. A strong correlation is also expected between histologic stain intensity and CT attenuation. EPIC-microCT shows the potential to be an effective tool in intervertebral disc analysis.

Ackowlegments

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